

PRISM: A Privileged Representation-Informed Synergistic Model for Transrectal Ultrasound-Only Prostate Segmentation

Anonymous Authors

Abstract—Accurate prostate segmentation in transrectal ultrasound (TRUS) is important for biopsy guidance, prostate-volume estimation, and downstream prostate cancer risk assessment. However, TRUS segmentation remains challenging because ultrasound images often suffer from weak soft-tissue contrast, acoustic shadowing, speckle noise, and ambiguous gland boundaries. Magnetic resonance imaging (MRI) provides clearer anatomical information, but its limited accessibility, higher cost, and additional examination burden make it unsuitable as a required input for routine TRUS-based workflows. We propose PRISM, a privileged representation-informed synergistic model for prompt-assisted TRUS prostate segmentation, where paired MRI provides training-stage privileged information while inference remains strictly TRUS-only. Built upon a lightweight medical foundation-model backbone, PRISM uses MRI as training-stage privileged anatomical information rather than as a test-time input. It introduces asymmetric privileged anatomical attention to transfer MRI-derived structural information into TRUS-centered representations, reliability-aware adaptive fusion to reduce adverse transfer under weak MRI-TRUS correspondence, and multi-bandwidth maximum mean discrepancy (MMD) alignment to improve latent compatibility without requiring pixel-level registration. On the public μ RegPro test set, PRISM achieves 92.95% DSC, 86.84% IoU, and 2.65 mm HD95; on 327 independent external TRUS frames, it achieves 87.42% DSC, 77.90% IoU, and 10.11 mm HD95. Since PRISM requires only TRUS input at inference, it remains lightweight and suitable for real-time prompt-assisted prostate segmentation without MRI acquisition. Anonymous code is available online.¹

Index Terms—Prostate segmentation, transrectal ultrasound, privileged learning, medical image segmentation.

I. INTRODUCTION

Prostate cancer is one of the most common malignancies in men, with approximately 1,466,680 new cases and 396,792 deaths reported worldwide in 2022 [1]. Early screening and pre-biopsy assessment remain clinically important [2], but serum prostate-specific antigen (PSA) testing has limited specificity and can lead to unnecessary invasive biopsies [3], [4]. Prostate volume (PV) estimation provides an important adjunct because prostate-specific antigen density (PSAD) depends directly on gland delineation [5], [6]. In practice, accurate PV estimation usually requires reliable prostate segmentation, but manual contouring is labor-intensive and observer-dependent [7], [8].

TRUS is attractive for prostate assessment because it is inexpensive, real-time, and compatible with biopsy guidance,

bedside examination, and intraoperative use. In these workflows, accurate gland delineation facilitates visual interpretation, treatment planning, and quantitative prostate assessment, while also supporting downstream prostate volume estimation and PSAD calculation when volumetric measurements are available. During TRUS-guided biopsy or routine ultrasound examination, the operator continuously adjusts the probe and selects diagnostically useful views while scanning. This workflow naturally supports a prompt-assisted segmentation paradigm: the sonographer provides coarse prostate localization, and the model performs the fine boundary delineation needed for volume estimation and visual verification. However, reliable TRUS prostate segmentation remains difficult due to speckle noise, acoustic shadowing, probe-dependent deformation, and weak tissue boundaries [9], [10]. MRI provides clearer soft-tissue anatomy, but its limited accessibility, higher cost, and additional examination burden prevent it from being a routine input for TRUS-based screening or intervention [11], [12]. This motivates a clinically practical learning paradigm in which MRI serves as privileged information during training, while the deployed model relies solely on TRUS during inference.

This objective naturally fits learning using privileged information (LUPI), where auxiliary information is available during training but omitted during testing [13], [14]. It differs from conventional multimodal segmentation, which often assumes that all modalities are present at inference. In our setting, MRI serves as training-stage privileged anatomical information, while the deployed model retains the acquisition cost, latency, and accessibility profile of a TRUS-only workflow. This formulation follows the clinical reality that MRI cannot be required for every TRUS screening examination.

Two limitations hinder the direct adoption of existing paradigms. First, TRUS prostate datasets remain limited, and the appearance of the gland varies substantially with probe pressure, acoustic shadowing, and acquisition angle. Medical foundation models such as MedSAM provide promptable segmentation priors learned from large-scale medical images [15], making them a natural starting point for real-time TRUS workflows in which coarse spatial prompts can be provided during scanning. Compared with conventional length-width measurements, prompt-assisted segmentation provides more complete gland contours for visual assessment and subsequent quantitative analysis. However, these generic foundation priors are not specifically optimized for prostate TRUS, where

¹<https://anonymous.4open.science/r/PRISM-07>

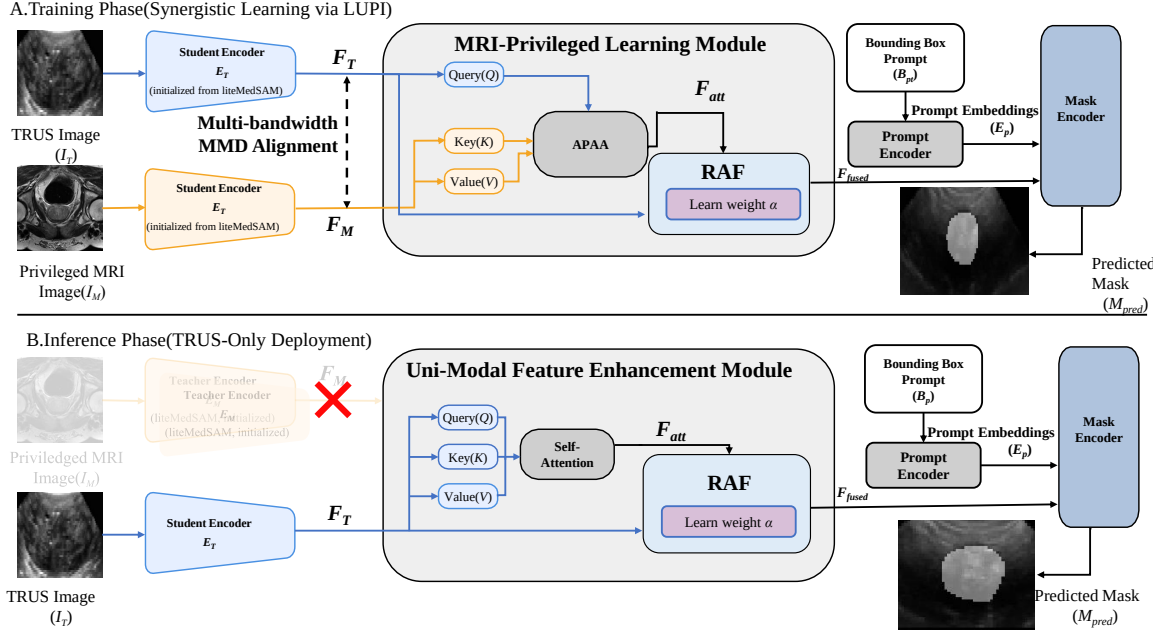


Fig. 1. Overview of PRISM. During training, MRI is used as privileged anatomical information to guide the TRUS branch through asymmetric privileged anatomical attention, reliability-aware adaptive fusion, and multi-bandwidth MMD alignment. During inference, the MRI branch is removed and PRISM performs prompt-assisted segmentation from TRUS alone.

weak soft-tissue contrast, speckle noise, and ambiguous gland boundaries remain prominent. A task-specific adaptation is therefore necessary. Second, MRI and TRUS are only weakly paired in realistic data. Probe-induced deformation and heterogeneous acquisition geometry make dense registration unreliable. Direct concatenation or symmetric fusion may therefore introduce mismatched MRI evidence, whereas treating MRI only as a generic regularizer underuses its anatomical value.

We propose PRISM, a privileged representation-informed synergistic model for prompt-assisted TRUS prostate segmentation. To the best of our knowledge, PRISM is the first framework that explicitly uses MRI as training-stage privileged anatomical information to improve TRUS-only prostate segmentation. Unlike conventional multimodal segmentation methods that require all modalities at inference, PRISM uses MRI only during training and deploys a lightweight TRUS-only model at test time. PRISM adapts a TinyViT-based LiteMedSAM backbone to preserve promptable medical foundation-model priors. It introduces three MRI-privileged learning components: asymmetric privileged anatomical attention (APAA), reliability-aware adaptive fusion (RAF), and multi-bandwidth MMD alignment. First, asymmetric privileged anatomical attention transfers MRI-derived structural information into TRUS-centered representations through target-oriented cross-attention. Second, reliability-aware adaptive fusion balances the original TRUS feature and the MRI-informed feature, reducing adverse transfer under weak MRI-TRUS correspondence. Third, multi-bandwidth MMD alignment improves latent compatibility between TRUS and MRI features without enforcing pixel-level registration. During inference, PRISM operates without the MRI branch and performs

prompt-assisted segmentation from TRUS alone.

Our contributions are summarized as follows:

- We propose, to the best of our knowledge, the first MRI-privileged learning framework for TRUS-only prostate segmentation, enabling MRI-informed representation learning without requiring MRI during inference.
- We develop a prompt-assisted privileged learning framework by adapting a LiteMedSAM backbone and introducing asymmetric privileged anatomical attention, reliability-aware adaptive fusion, and multi-bandwidth latent distribution alignment to effectively transfer MRI-derived anatomical knowledge into TRUS representations during training.
- We validate PRISM on 3,507 paired public training/validation slices, 366 internal test slices, and 327 independent external TRUS frames, demonstrating strong segmentation performance and robust cross-dataset generalization.

II. METHODS

A. MRI-Privileged Learning for TRUS-Only Inference

Let $x \in \mathcal{X}$ denote a TRUS image, $x^* \in \mathcal{X}^*$ denote its paired but not pixel-wise registered MRI image, b denote a coarse bounding-box prompt, and $y \in \mathcal{Y}$ denote the prostate mask. The training set is

$$\mathcal{S}_{\text{train}} = \{(x_i, x_i^*, b_i, y_i)\}_{i=1}^N. \quad (1)$$

TRUS is the target modality available at both training and inference, whereas MRI is the privileged modality available

only during training. The deployed model predicts from TRUS and its prompt alone:

$$\hat{y} = f_{\theta}(x, b). \quad (2)$$

This setting differs from conventional multimodal segmentation, where all modalities are commonly treated as inference inputs. In PRISM, MRI is not used to form a test-time multimodal input. Instead, it provides training-stage anatomical information to improve the TRUS representation. This formulation is well suited to prostate TRUS segmentation because ultrasound images often suffer from acoustic shadowing, speckle noise, weak edge contrast, and probe-dependent appearance variation, whereas MRI provides more stable soft-tissue structural information.

PRISM therefore adopts an asymmetric privileged learning scheme. During training, the MRI branch informs the TRUS branch through representation-level feature interaction. The privileged signal is transferred in the latent space rather than through dense spatial supervision, avoiding the need for pixel-wise MRI-TRUS registration. During inference, MRI is removed and prediction is performed using only the TRUS image and the bounding-box prompt.

B. PRISM Architecture

1) Modality-Specific Encoding via Foundation Priors:

Training a high-capacity segmentation model from scratch for TRUS is challenging because public datasets with dense annotations remain limited. To improve data efficiency and training stability, PRISM is initialized from the lightweight LiteMedSAM backbone. Specifically, we use TinyViT [16] as the image encoder instead of the original ViT-Base backbone used in SAM-style foundation models [17]. This lightweight design reduces computational burden and inference latency, which is important for real-time TRUS examination. In our setting, increasing the encoder capacity substantially raises computational cost, while the accuracy gain remains limited relative to the additional inference burden. Therefore, PRISM adopts a lightweight encoder to support efficient prompt-assisted TRUS segmentation.

TRUS and MRI exhibit different texture statistics, noise sources, and boundary appearances. We therefore use two modality-specific encoders during training: a TRUS encoder $E_T(\cdot)$ and an MRI encoder $E_M(\cdot)$. Given a TRUS image I_T and its paired MRI image I_M , the two branches produce

$$F_T = E_T(I_T), \quad F_M = E_M(I_M), \quad (3)$$

where F_T is the target-modality feature associated with the deployed TRUS pathway and F_M is the privileged anatomical feature available only during training. The MedSAM prompt encoder $B(\cdot)$ maps the bounding-box prompt b into prompt embeddings, which are passed with image features to the mask decoder $D(\cdot)$.

2) *Asymmetric Privileged Anatomical Attention*: The core interaction module of PRISM is an asymmetric privileged anatomical attention block. Its design is motivated by the complementary roles of the two modalities: TRUS is the target

modality used for deployment, whereas MRI provides clearer anatomical context during training. Instead of concatenating TRUS and MRI features or treating them symmetrically, PRISM uses the TRUS feature as the query source and the privileged MRI feature as the key-value source.

Let $\mathcal{A}(F_q, F_{kv})$ denote a multi-head attention operator that uses F_q to query structural information from F_{kv} :

$$\mathcal{A}(F_q, F_{kv}) = \phi_O \left[\mathcal{S} \left(\frac{\phi_Q(F_q) \phi_K(F_{kv})^\top}{\sqrt{d}} \right) \phi_V(F_{kv}) \right], \quad (4)$$

where $\phi_Q(\cdot)$, $\phi_K(\cdot)$, $\phi_V(\cdot)$, and $\phi_O(\cdot)$ are learnable projections, $\mathcal{S}(\cdot)$ denotes the Softmax operation, and d is the feature dimension of each attention head. During training, MRI-informed TRUS features are obtained as

$$F_{\text{MRI} \rightarrow \text{TRUS}} = \text{Norm}(F_T + \mathcal{A}(F_T, F_M)). \quad (5)$$

This residual formulation preserves the original TRUS representation while injecting MRI-derived structural cues through feature-level attention.

The asymmetry is deliberate. The output is not a generic multimodal embedding; it is an MRI-informed TRUS representation. Since TRUS is the only imaging modality available at deployment, the privileged MRI branch is used to shape the target-modality feature rather than to form a test-time multimodal input.

For prostate TRUS segmentation, this design is clinically motivated. The prostate boundary in ultrasound is often locally ambiguous due to speckle noise, acoustic shadowing, and weak soft-tissue contrast. MRI provides more stable soft-tissue structure during training, and the privileged attention block encourages the TRUS feature to encode anatomical cues that are difficult to infer from ultrasound appearance alone, without requiring explicit pixel-wise MRI-TRUS registration.

3) *Reliability-Aware Adaptive Fusion*: Because MRI and TRUS are weakly paired rather than strictly registered, the MRI-informed feature is not uniformly reliable. Directly replacing the original TRUS feature with $F_{\text{MRI} \rightarrow \text{TRUS}}$ may introduce negative transfer when cross-modal correspondence is imperfect. PRISM therefore adopts a reliability-aware adaptive fusion (RAF) module to balance the original TRUS representation and the MRI-informed enhancement.

Given the original TRUS feature F_T and the MRI-informed feature $F_{\text{MRI} \rightarrow \text{TRUS}}$, a lightweight reliability estimation module predicts a confidence coefficient

$$\alpha = \sigma(h(F_T, F_{\text{MRI} \rightarrow \text{TRUS}})), \quad (6)$$

where $h(\cdot)$ denotes the reliability estimation function and $\sigma(\cdot)$ is the sigmoid activation. The resulting gate adaptively regulates the contribution of privileged MRI information:

$$F_{\text{cat}} = [(1 - \alpha) \odot F_T, \alpha \odot F_{\text{MRI} \rightarrow \text{TRUS}}]. \quad (7)$$

The concatenated feature is then projected back to the decoder dimension through a 1×1 fusion layer,

$$F_{\text{fuse}} = \psi(F_{\text{cat}}), \quad (8)$$

where $\psi(\cdot)$ denotes the fusion projection. In this way, RAF preserves the original TRUS representation while selectively incorporating MRI-derived anatomical information according to its estimated reliability.

The training-stage prediction is obtained by the prompt-conditioned mask decoder:

$$\hat{y}_{\text{train}} = D(F_{\text{fuse}}, B(b)). \quad (9)$$

4) *TRUS-Only Inference*: At inference time, the MRI encoder is removed and no MRI image is required. Given a TRUS image, the TRUS encoder extracts

$$F_T = E_T(I_T). \quad (10)$$

To retain the feature-processing pathway optimized during MRI-privileged training, the trained interaction module can be operated in a unimodal form by using TRUS features as query, key, and value:

$$F_{\text{TRUS} \rightarrow \text{TRUS}} = \text{Norm}(F_T + \mathcal{A}(F_T, F_T)). \quad (11)$$

The adaptive fusion module then combines the original TRUS feature and the self-enhanced TRUS feature:

$$F_{\text{infer}} = \psi([(1 - \alpha_T) \odot F_T, \alpha_T \odot F_{\text{TRUS} \rightarrow \text{TRUS}}]), \quad (12)$$

where α_T is computed from TRUS-only descriptors using the same gating form as in training. The final prediction is

$$\hat{y}_{\text{infer}} = D(F_{\text{infer}}, B(b)). \quad (13)$$

Thus, PRISM keeps the deployed model TRUS-only while retaining a lightweight feature interaction and prompt-conditioned decoding pathway that has been optimized under MRI-privileged training.

C. Training Objective

The optimization objective combines prompt-guided segmentation supervision with multi-bandwidth latent distribution alignment:

$$\mathcal{L} = \lambda_d \mathcal{L}_{\text{dice}} + \lambda_c \mathcal{L}_{\text{ce}} + \lambda_i \mathcal{L}_{\text{iou}} + \lambda_m \mathcal{L}_{\text{mmd}}. \quad (14)$$

Dice loss and binary cross-entropy supervise mask overlap and pixel-wise classification. The IoU regression loss supervises the predicted mask quality score:

$$\mathcal{L}_{\text{iou}} = \|\hat{q} - \text{IoU}(\mathbb{1}[\sigma(\hat{y}) > 0.5], y)\|_2^2, \quad (15)$$

where $\hat{q}$ is the predicted IoU score from the mask decoder.

1) *Multi-Bandwidth Latent Distribution Alignment*: In addition to feature interaction, PRISM regularizes the TRUS and MRI latent spaces using a multi-bandwidth MMD alignment loss [18]. The purpose is not to enforce spatial registration, but to reduce the distribution gap between modality-specific features so that MRI-derived cues can be transferred more stably during training.

Given mini-batch TRUS features $\{F_T^i\}_{i=1}^n$ and MRI features $\{F_M^j\}_{j=1}^m$, the MMD loss is defined as

$$\begin{aligned} \mathcal{L}_{\text{mmd}} = & \frac{1}{n^2} \sum_{i,j} k(F_T^i, F_T^j) + \frac{1}{m^2} \sum_{i,j} k(F_M^i, F_M^j) \\ & - \frac{2}{nm} \sum_{i,j} k(F_T^i, F_M^j). \end{aligned} \quad (16)$$

Instead of using a single Gaussian kernel, PRISM uses a multi-bandwidth kernel:

$$k(u, v) = \sum_{\ell=1}^L \exp\left(-\frac{\|u - v\|_2^2}{\sigma_\ell}\right), \quad (17)$$

where $\{\sigma_\ell\}_{\ell=1}^L$ denotes a set of bandwidths arranged in a geometric progression. This multi-bandwidth design allows the alignment term to capture distribution discrepancies at different feature-distance scales. Large-bandwidth kernels provide coarse global alignment, whereas small-bandwidth kernels are more sensitive to local feature discrepancies.

For prostate TRUS segmentation, this distribution-level constraint is preferable to registration-based supervision. MRI and TRUS provide related gland anatomy but are not guaranteed to be pixel-wise aligned due to probe-induced deformation and acquisition differences. Multi-bandwidth MMD therefore improves latent compatibility between the two modalities without imposing unrealistic geometric correspondence.

III. EXPERIMENTS

A. Datasets and Implementation

We evaluated PRISM on two datasets (Table I). The public μRegPro dataset [19] provides 3,507 paired MRI-TRUS training/validation slices and 366 internal test slices. To assess external generalization under the clinical workflow, we additionally collected an independent TRUS dataset comprising 327 frames acquired during real-time prostate ultrasound assessment using a BK5000 ultrasound system. This dataset consists of standalone TRUS images without paired MRI or standardized geometric metadata. To simulate the prompt-assisted workflow, experienced clinicians annotated coarse prostate bounding boxes on these TRUS frames, which were used as box prompts during evaluation. Institutional review board approval was obtained.

Each 3D volume was decomposed into 2D slices. Slices containing prostate labels were retained, and the original

TABLE I
DATASETS AND EVALUATION PARTITIONS.

ROLE	MODALITY	2D IMAGES
μRegPro		
Train/Val	Paired MRI+TRUS	3507
Internal test	Paired MRI+TRUS	366
Self-collected external cohort		
External test	TRUS	327

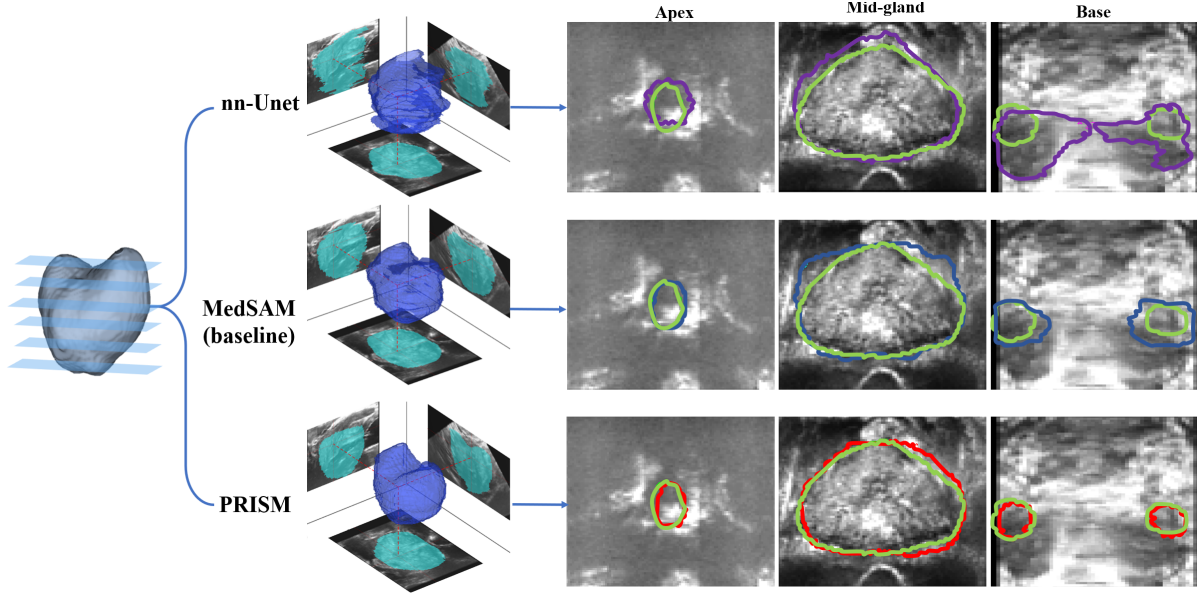


Fig. 2. Representative prostate segmentation results across apex, mid-gland, and base. PRISM yields smoother and more anatomically valid boundaries than MedSAM and nnU-Net, especially in shadowed or low-contrast regions.

spacing information was saved for physical-distance evaluation. Images were converted to three-channel inputs, intensity-normalized, resized, and padded to 256×256 . Masks were resized with nearest-neighbor interpolation. Index-matched MRI-TRUS pairs were used for privileged training, while MRI was not used at test time.

Bounding-box prompts were generated from prostate masks during training and internal testing by expanding the tight bounding box by 5 pixels and clipping it to the image boundary. On the external TRUS cohort, clinician-annotated coarse boxes were used as evaluation prompts. In all settings, the box prompt provided only coarse localization, and the model predicted the final prostate boundary.

All models used the same data partitions, preprocessing pipeline, and evaluation protocol. PRISM was trained for 20 epochs with AdamW, an initial learning rate of 1×10^{-4} , weight decay of 0.01 on an NVIDIA RTX 5070 Ti. For comparison methods, we followed their official open-source implementations, released checkpoints or pretrained weights when available, and recommended training protocols to fine-tune them on the μ RegPro training partition. All methods were evaluated on the same internal test partition and with the same evaluation protocol.

B. Evaluation Metrics and Baselines

All quantitative results are reported under a 2D evaluation protocol. We use Dice similarity coefficient (DSC), Intersection over Union (IoU), and 95% Hausdorff distance (HD95) as evaluation metrics. DSC and IoU measure region overlap, whereas HD95 measures boundary error. For a prediction P and ground truth G ,

$$\text{DSC} = \frac{2|P \cap G|}{|P| + |G|}, \quad \text{IoU} = \frac{|P \cap G|}{|P \cup G|}. \quad (18)$$

HD95 is computed as the 95th percentile of bidirectional surface distances. All metrics are computed for each 2D slice or frame and then averaged over the test set.

We compare PRISM with four categories of baselines. The first category includes fully automatic segmentation models, including 3D U-Net [20], UNETR++ [21], nnU-Net [22], and SAMed [23]. The second category includes 2D box-prompted SAM-style models, including LiteMedSAM [15] and SAM-Med2D [24]. The third category includes volumetric or interactive prompt-guided models, including SegVol [25] and nnInteractive [26], representing prompt-based interactive medical image segmentation settings [27]. The fourth category includes modality-missing multimodal models, including RFNet [28] and mmFormer [29], which are trained with paired TRUS/MRI data and evaluated using TRUS only; this setting is related to privileged unimodal learning and incomplete multimodal segmentation [30], [31]. All comparison methods were reproduced using publicly available implementations and checkpoints. Methods supporting task-specific adaptation were fine-tuned on the μ RegPro training set, whereas the remaining methods were evaluated using their official pretrained weights. Prompt-guided methods were evaluated using the same box-prompt protocol when applicable. Since the external dataset consists of non-contiguous 2D TRUS frames, external results are reported only for methods supporting valid 2D inference.

IV. RESULTS

A. Comparison on Internal and External Datasets

Table II summarizes the quantitative comparison on the internal μ RegPro test set and the independent external TRUS dataset. On the internal μ RegPro benchmark, PRISM achieves the best overall performance among all compared methods, reaching 92.95% DSC, 86.84% IoU, and 2.65 mm

TABLE II
QUANTITATIVE COMPARISON ON THE INTERNAL μ REGPRO TEST SET AND THE EXTERNAL TEST SET.

#	Method	Params (M)	μ RegPro			External Dataset		
			DSC (%) $\uparrow$	IoU (%) $\uparrow$	HD95 (mm) $\downarrow$	DSC (%) $\uparrow$	IoU (%) $\uparrow$	HD95 (mm) $\downarrow$
1	3D U-Net	4.81	77.39	66.78	10.67	–	–	–
2	UNETR++	42.95	77.52	67.21	11.56	–	–	–
3	nnU-Net	24.27	84.45	75.95	8.85	52.72	39.95	83.76
4	SAMed	91.35	75.07	65.76	37.10	66.53	54.22	39.18
5	LiteMedSAM	9.79	88.82	80.29	3.79	83.08	71.49	15.84
6	SAM-Med2D	271.24	85.78	76.82	13.66	44.43	32.14	99.92
7	SegVol	180.89	82.79	73.81	17.89	–	–	–
8	nnInteractive	383.50	78.32	65.37	25.73	76.99	63.97	62.30
9	RFNet	8.69	76.46	64.68	36.63	–	–	–
10	mmFormer	36.65	69.94	56.67	72.74	–	–	–
11	PRISM	15.92	92.95	86.84	2.65	87.42	77.90	10.11

HD95. Compared with the task-adapted LiteMedSAM baseline, PRISM improves DSC by more than 4 percentage points and IoU by over 6 percentage points, while reducing HD95 from 3.79 mm to 2.65 mm. These gains demonstrate that simply adapting a foundation model to TRUS is insufficient, whereas incorporating MRI as privileged supervision substantially improves both region overlap and boundary localization.

Compared with conventional automatic segmentation networks, PRISM consistently achieves superior performance. Relative to nnU-Net, PRISM improves DSC by more than 8 percentage points while reducing HD95 by approximately 70%. Similar trends are observed against 3D U-Net and UNETR++, indicating that the proposed privileged representation learning strategy is more effective than purely volumetric modeling under challenging TRUS appearance variations.

PRISM also consistently outperforms other prompt-guided and foundation-model-based methods. Compared with SAM-Med2D and nnInteractive, PRISM produces higher overlap accuracy together with markedly lower boundary error, despite requiring considerably fewer inference parameters. Moreover, we adapted representative incomplete multimodal segmentation methods, RFNet and mmFormer, to our paired MRI-TRUS prostate segmentation setting and evaluated them under the same TRUS-only inference protocol. Despite leveraging paired MRI during training, both methods perform substantially worse than PRISM when MRI is unavailable at test time. These results indicate that an MRI-privileged learning strategy is better suited to the MRI-TRUS setting than directly applying existing incomplete multimodal segmentation methods.

On the independent external dataset, PRISM again achieves the best overall performance among all methods supporting 2D inference, reaching 87.42% DSC, 77.90% IoU, and 10.11 mm HD95. Compared with LiteMedSAM, PRISM consistently improves both overlap accuracy and boundary localization, indicating that the proposed privileged learning strategy generalizes beyond the source training distribution. The advantage becomes even more evident when compared with generic segmentation models. nnU-Net and SAM-Med2D experience substantial performance degradation under cross-center domain

shift, whereas PRISM maintains stable segmentation accuracy. These observations indicate that MRI-guided representation learning improves the robustness of TRUS feature extraction rather than merely fitting the training distribution.

Qualitatively, Fig. 2 shows that baseline methods often leak into adjacent tissues or produce fragmented contours in low-contrast regions. PRISM produces smoother and more anatomically plausible prostate boundaries while preserving high region-level agreement. This visual observation is consistent with PRISM’s high region-level accuracy on the internal test set and its strong boundary robustness on the external cohort.

B. Ablation Analysis

Table III evaluates the contribution of each component in PRISM. Directly applying the pretrained LiteMedSAM model (EXP0) achieves only 70.07% DSC, whereas task-specific fine-tuning (EXP1) improves the performance to 88.82% DSC and 80.29% IoU. This confirms that, despite strong foundation-model priors, dedicated adaptation remains essential for prostate TRUS segmentation because of its unique imaging characteristics.

To investigate the privileged learning strategy, we progressively introduce the proposed modules. Adding asymmetric privileged anatomical attention alone (EXP2) slightly degrades the baseline performance, indicating that directly transferring MRI-derived information is insufficient under weakly paired MRI-TRUS correspondence and may introduce unreliable cross-modal cues. After incorporating reliability-aware adaptive fusion (EXP3), the model achieves a marked improvement to 91.62% DSC and 84.99% IoU, while reducing HD95 to 2.39 mm. This demonstrates that selectively regulating the contribution of privileged MRI features is critical for exploiting complementary anatomical information while suppressing negative transfer.

Finally, introducing multi-bandwidth MMD alignment (EXP4) further improves the overall overlap accuracy, achieving the best DSC (92.95%) and IoU (86.84%). Although EXP3 attains the lowest HD95, the complete model provides

TABLE III
ABLATION STUDY OF THE PROPOSED COMPONENTS ON μ REGPRO.

Method	DSC (%) $\uparrow$	IoU (%) $\uparrow$	HD95 (mm) $\downarrow$
EXP0 (Pretrained LiteMedSAM)	70.07	55.90	7.64
EXP1 (Task-tuned LiteMedSAM)	88.82	80.29	3.79
EXP2 (+ APAA)	87.00	78.37	3.89
EXP3 (+APAA + RAF)	91.62	84.99	2.39
EXP4 (+APAA + RAF + MMD, full PRISM)	92.95	86.84	2.65

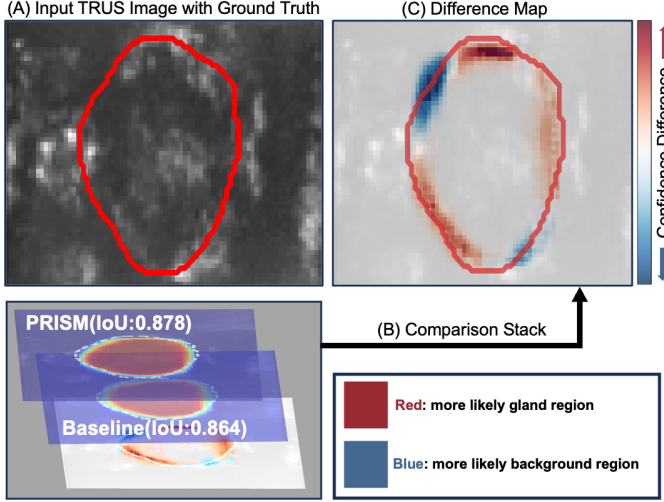


Fig. 3. Effect of MRI privileged priors on prediction confidence. Difference maps indicate that PRISM increases confidence inside the gland while suppressing ambiguous background responses near weak boundaries.

the strongest overall segmentation performance. These results suggest that APAA and RAF primarily enhance local boundary representation, whereas MMD alignment further improves cross-modal feature compatibility, leading to more consistent region-level segmentation.

C. Clinical Deployment, Prompt Robustness, and Limitations

To further investigate how privileged learning influences prediction behavior, Fig. 3 visualizes the confidence difference between LiteMedSAM and PRISM. The largest confidence gains consistently occur around ambiguous prostate boundaries rather than homogeneous interior regions, indicating that MRI-privileged supervision primarily improves anatomical boundary discrimination instead of simply increasing prediction confidence globally. These visual observations agree with the substantial reduction in HD95 reported in Table II.

Prompt-guided models must also tolerate variations in user interaction. To simulate this variability, we evaluated PRISM under two types of bounding-box perturbations: expansion, which mimics loose framing, and translation, which mimics off-center placement. As shown in Fig. 4, PRISM maintains DSC above 80% within a moderate perturbation range. This indicates that the bounding box mainly acts as a coarse localization cue rather than a strict boundary surrogate. Therefore, the model does not require highly precise box placement, which may reduce the operational burden during real-time TRUS examination.

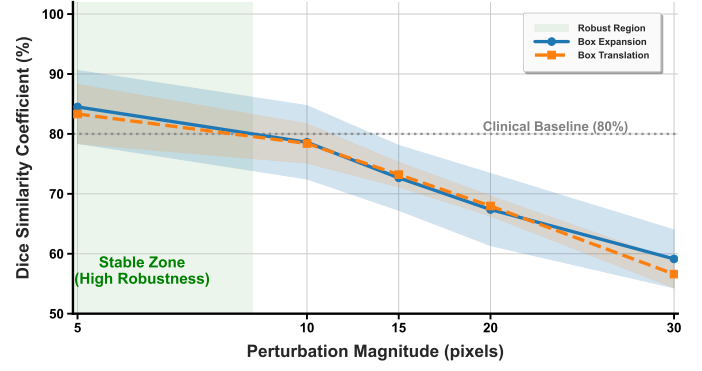


Fig. 4. Sensitivity analysis to bounding-box prompt perturbations. Box expansion simulates loose framing; box translation simulates off-center framing. Shaded bands indicate 95% confidence intervals.

These properties are clinically relevant for prompt-assisted TRUS examination and biopsy workflows, where robust prostate delineation facilitates consistent anatomical localization and visual assessment. By combining box-guided interaction, boundary confidence visualization, and MRI-privileged representation learning, PRISM provides a practical deployment paradigm: MRI serves only as privileged anatomical information during training, while inference requires only a TRUS image and a coarse box prompt. This asymmetric train-test design enables the model to benefit from MRI-derived anatomical knowledge without increasing deployment complexity. At inference, the MRI branch is completely removed, resulting in a lightweight model with 15.92M parameters that processes each 2D slice in 14.61 ms on an RTX 5070 Ti, supporting real-time clinical deployment.

Several limitations remain. First, PRISM currently performs slice-wise 2D inference because both the available foundation model and the prompt-guided training pipeline are inherently 2D. Extending the framework to lightweight 2.5D or 3D architectures is a promising direction for improving inter-slice anatomical consistency while preserving efficient deployment. Second, although the box-perturbation experiment provides a controlled evaluation of prompt robustness, prospective studies with real user interactions are needed to quantify inter-operator variability in clinical use [32]. Third, the external validation cohort remains limited in scale. Larger multi-center and multi-vendor studies are needed to further evaluate the generalizability of MRI-privileged learning across different acquisition protocols and clinical environments.

V. CONCLUSION

In summary, PRISM introduces an MRI-privileged learning framework for prompt-assisted TRUS prostate segmentation. It uses MRI only during training to improve TRUS representations through asymmetric privileged anatomical attention, reliability-aware adaptive fusion, and multi-bandwidth MMD alignment, while requiring only TRUS and a box prompt at inference. On the internal μ RegPro test set, PRISM achieves 92.95% DSC, 86.84% IoU, and 2.65 mm HD95; on the independent external TRUS dataset, it achieves 87.42% DSC,

77.90% IoU, and 10.11 mm HD95. These results demonstrate that privileged anatomical supervision offers a practical alternative to test-time multimodal fusion, providing an effective paradigm for deploying foundation models in modality-limited clinical imaging scenarios.

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